CHAPTER 2

LINEAR, TIME INVARIANT SYSTEMS

Two of the basic system properties presented in the previous chapter, namely linearity and time invariance play an important role in signal and system analysis. This is because many physical phenomena can be modeled as linear, time invariant (LTI) systems rather satisfactorily and, also because there are powerful mathematical tools for the analysis of such systems.

If the input to a LTI system consists of a linear combination of signals

$$x = a_1 x_{1+} a_2 x_{2+} a_3 x_{3+} \ldots\ldots$$

then, by superposition, the output is

$$y = a_1 y_{1+} a_2 y_{2+} a_3 y_{3+} \ldots\ldots$$

where y_k is the response to x_k, k=1,2,3,... and y, and x, represent discrete time or continuous time signals as the case may be. Therefore, if we can represent the input in terms of a set of basic signals, we can use superposition to compute the output.

2.1. THE REPRESENTATION OF SIGNALS IN TERMS OF IMPULSES

We will first examine the discrete time case. Consider the discrete time signal $x[n]$ of Figure 2.1.1 (a). At the rest of the figure, certain weighted and shifted unit impulses are given. The weight of each is equal to the value of $x[n]$ at the particular time instant at which the unit impulse is located. Thus, the sum of the four sequences in the figure,

$$x = x[-1]\delta[n+1] + x[0]\delta[n] + x1\delta[n-1] + x[2]\delta[n-2] \qquad (2.1.1)$$

will yield $x[n]$ for -1<n<2. Including more shifted and weighted impulses one can write

$$x(t) = \cdots + x[-1]\delta[n+1] + x[0]\delta[n] + x1\delta[n-1] + x[2]\delta[n-2] + \cdots (2.1.2)$$

$$= \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]\,. \qquad (2.1.3)$$

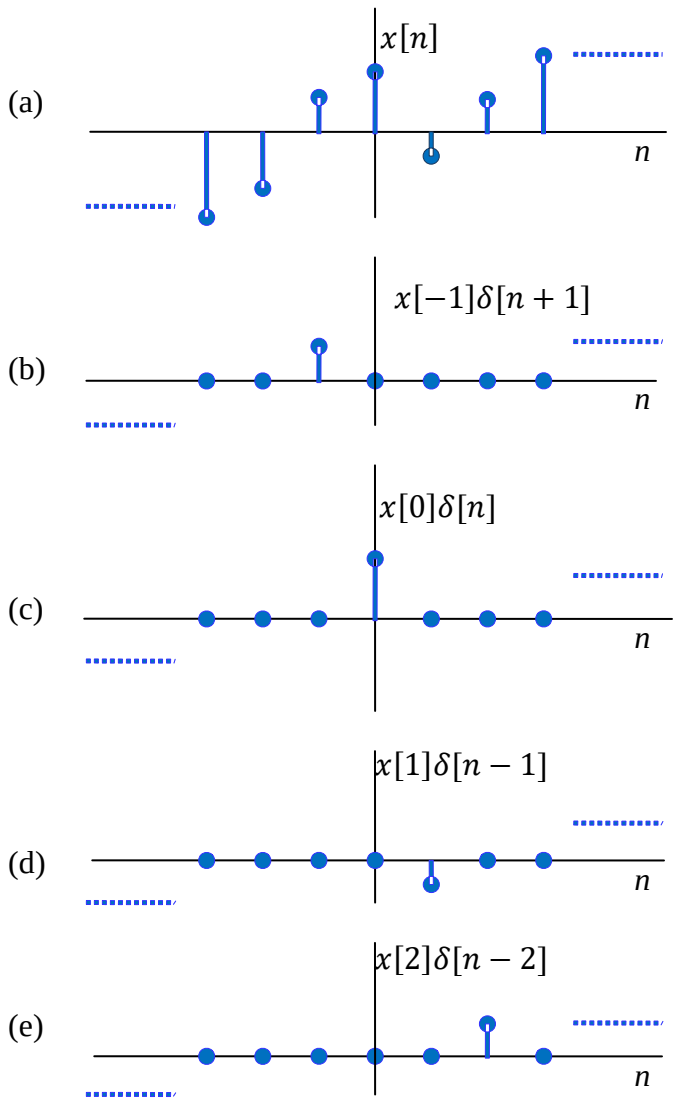

Fig.2.1.1. Decomposition of a discrete-time signal into a weighted sum of shifted impulses.

Therefore, any arbitrary sequence can be represented as a linear combination of shifted impulses $\delta[n-k]$. For example, if $x[n]=u[n]$, -since $u[k]=0$ for k<0 and $u[k]=1$ for k≥0-, one can write

$$u[n] = \sum_{k=-\infty}^{\infty} \delta[n-k] \quad . \tag{2.1.4}$$

(2.1.3) is known as the "sifting property" of the discrete time unit impulse.

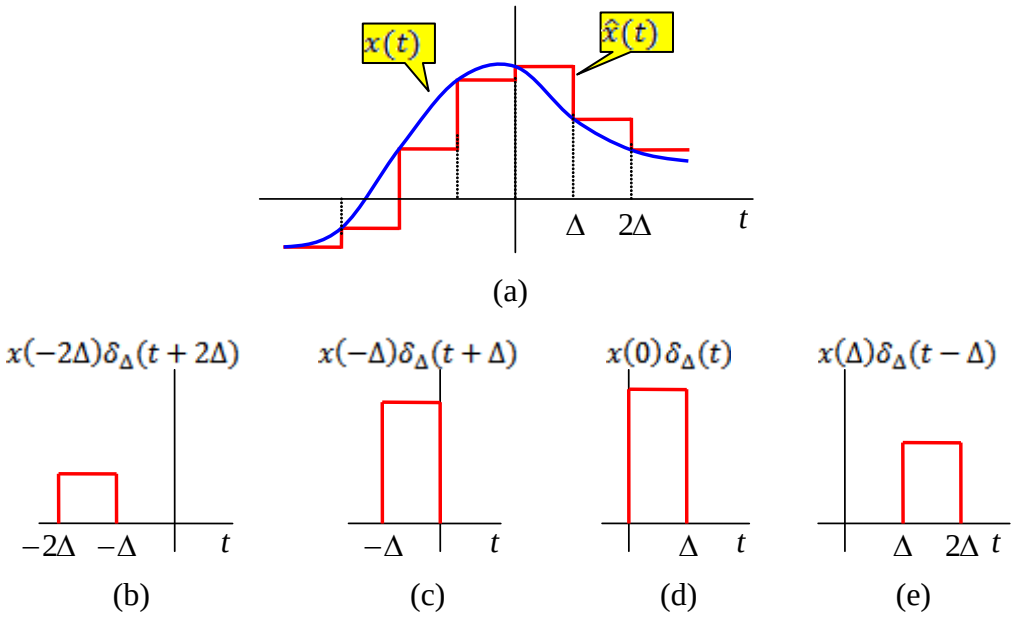

Fig.2.1.2. Staircase approximation to a continuous-time signal and its decomposition

A continuous time signal can be represented in terms of unit impulse functions in a similar manner. Consider the staircase approximation $\hat{x}(t)$, to the continuous time signal $x(t)$ as shown in Figure 2.1.2(a). This approximation can be written as linear combination of delayed pulses as shown in the rest of the same figure.

If we define

$$\delta(t) = \begin{cases} 1/\Delta & 0 < t < \Delta \\ 0 & \text{otherwise} \end{cases}, \tag{2.1.5}$$

then, since $\Delta\delta_\Delta(t)$ has unit amplitude,

$$\hat{x}[t] = \sum_{k=-\infty}^{\infty} x(k\Delta)\delta_\Delta(t-k\Delta).\Delta \quad . \tag{2.1.6}$$

As we reduce Δ, the approximation becomes better. Therefore

$$x(t) = \lim_{\Delta\to 0} \hat{x}(t) = \lim_{\Delta\to 0} \sum_{k=-\infty}^{\infty} x[k\Delta]\delta(t-k\Delta).\Delta \quad . \tag{2.1.7}$$

As Δ→0, the summation becomes an integral and $\delta_\Delta(t)$ approaches $\delta(t)$. Therefore

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau \quad . \tag{2.1.8}$$

As in the discrete-time case, (2.1.8) is known as the sifting property of the unit impulse. Note that for the particular case $x(t) = u(t)$ we get

$$\boldsymbol{u(t) = \int_{-\infty}^{\infty} u(\tau)\,\delta(t-\tau)d\tau = \int_{0}^{\infty} \delta(t-\tau)d\tau} \ . \tag{2.1.9}$$

2.2. DISCRETE-TIME LTI SYSTEMS; THE CONVOLUTION SUM

Consider a LTI discrete-time system with $x[n]$ as the input sequence. If we express $x[n]$ as a linear combination of shifted unit impulses as

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \ , \tag{2.2.1}$$

the response to $x[n]$ can be calculated using the superposition property. If the response to $\delta[n\text{-}k]$ is $h_k[n]$ then the total response is

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h_k[n] \, . \tag{2.2.2}$$

because of linearity. On the other hand, because of time invariance

$$\delta[n-k] \rightarrow h_k[n] = h[n-k] \tag{2.2.3}$$

where $h[n]$ is the response to $\delta[n]$ and is known as the <u>unit impulse response</u>. Thus, the response is given as

$$y[n] = \textstyle\sum_{k=-\infty}^{\infty} x[k]h[n-k] \tag{2.2.4}$$

A graphical interpretation of this result is given in Figure 2.2.1. The finite sequence $x[n]$ shown in Figure 2.2.1(a) stimulates a system whose unit impulse response is shown in Figure 2.2.1(b). Responses to individual "shifted impulse" type components are given in part (c) and together they form the complete response $y[n]$ shown in part (d) of the same figure.

This result is known as the "convolution sum" and the operation on the right hand side is known as the "convolution of two sequences $x[n]$ and $h[n]$" which will be represented symbolically as

$$y[n] = x[n] * h[n]. \tag{2.2.5}$$

Note that the convolution operation is commutative. (The reader is urged to prove this fact as an exercise.)

In short, if the unit impulse response $h[n]$ of a LTI discrete-time system is known, then the response to an arbitrary input sequence $x[n]$ can be found as

$$x[n] \rightarrow y[n] = x[n] * h[n]. \tag{2.2.6}$$

Here we see that a LTI discrete-time system is completely characterized by its impulse response.

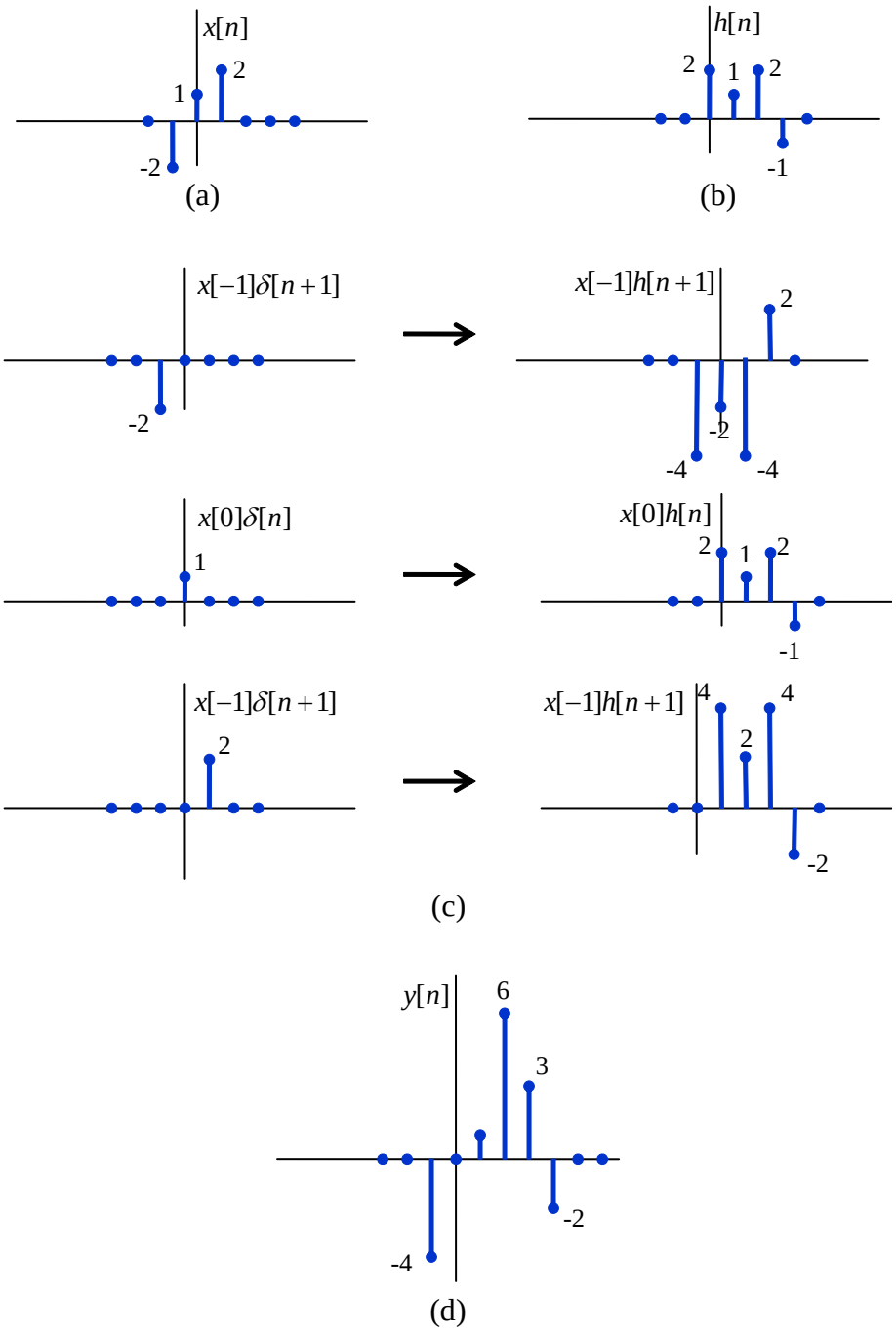

Fig.2.2.1. Formation of the response in a discrete-time LTI system

According to (2.2.4), the value $y[n]$ of the output at any instant ***n***,consists of the sum over all ***k*** values of the numbers $x[k]h[n-k]$. The term $x[k]$ can be obtained from $h[k]$ simply by reflection about the horizontal axis and a shift of n units (to the right if n>0). Then the product $x[k]h[n-k]$ is obtained by multiplying $x[k]$ and $h[n-k]$. This procedure is illustrated in Figure 2.2.2. By repeated application of the same procedure for different ***n***, the sequence $y[n]$ will be found. Hence, the convolution operation can be viewed as a repeated application of a "fold-shift-multiply-add" routine. This point is illustrated further by the following example.

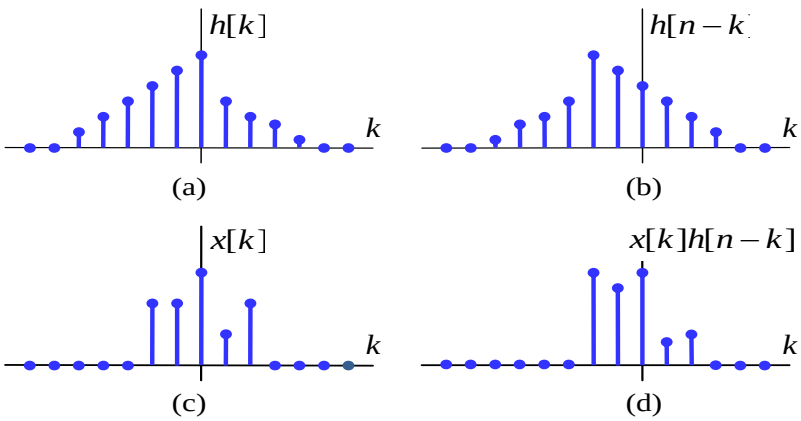

Fig.2.2.2. Interpretation of (2.2.4): the signal $h[n]$ (a), is reflected and shifted by n (here shown for n=2)(b), multiplied by $x[n]$ (c-d)

Example 2.2.1.

Let the impulse response h[n] of and the input sequence x[n] to a LTI system be as shown in Figure 2.2.3(a) and (b). Figure 2.2.3(c)-(g) depict h[n-k] for n=-l,0,l,2 and 3 respectively. The product of these sequences by x[n] are given in part (h)-(l) and finally, the response y[n] is as shown in part (m) of the same figure.

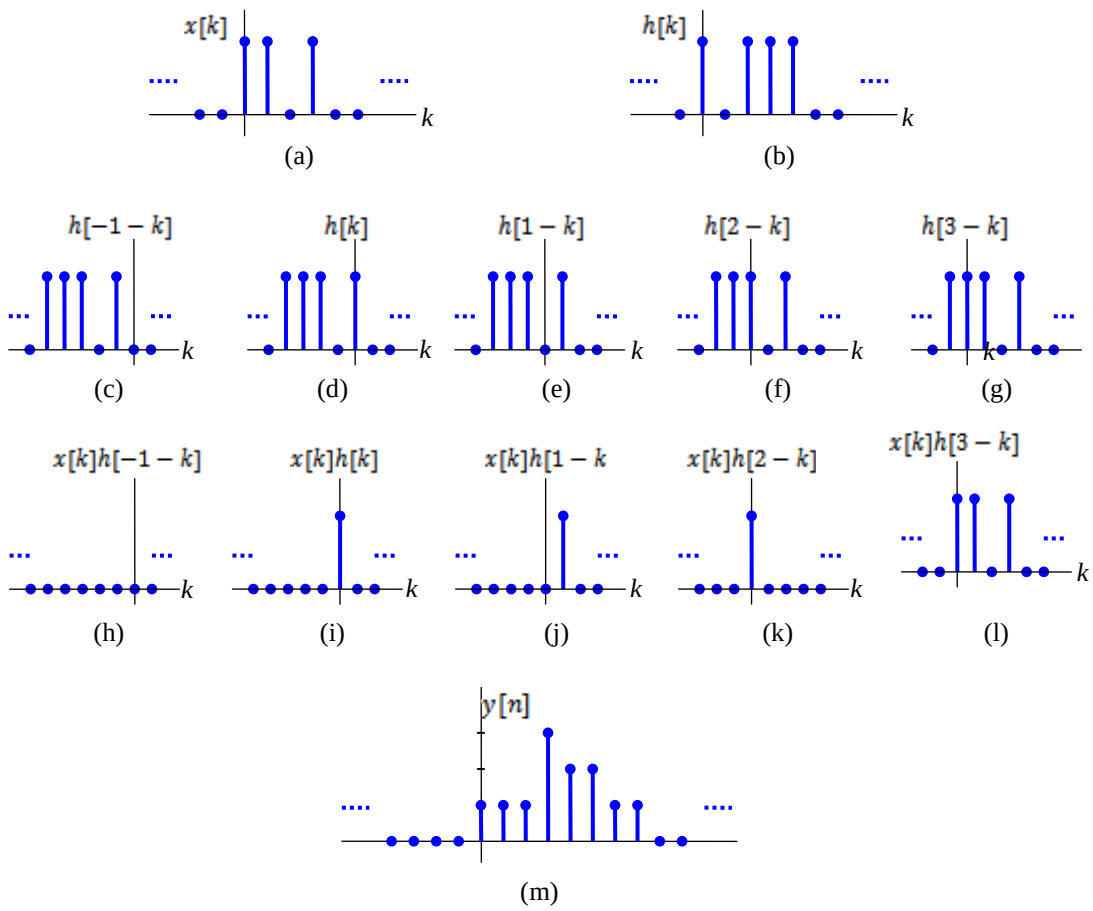

Fig.2.2.3. Graphical interpretation of the convolution performed in Example 2.2.1

Example 2.2.2.

Let us determine the convolution $x[n] * h[n]$ where

$$x[n] = \begin{cases} 2^n & n > 0 \\ 0 & n \le 0 \end{cases}$$

and

$$h[n] = \begin{cases} n & n > 0 \\ 0 & n \le 0 \end{cases}.$$

Figure 2.2.4(a-c) depict $x[k]$, $h[k]$ and $h[-k]$ respectively, for a positive value of n, $h[-k]$ will be shifted to the right by n units. There are two different intervals of n for which the procedure assumes different characters.

Interval 1: For n<l, the product x[k]h [n-k] is always zero. Therefore y[n] =0 for n<2.

Interval 2: For n>l x[k]h [n-k] will be non-zero. Thus

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=1}^{\infty} 2^k h[n-k]$$
$$= \sum_{k=1}^{n-1} 2^k (n-k) = 2(2^n - n - 1)$$

The resulting convolution is depicted in Figure 2.2.4(d).

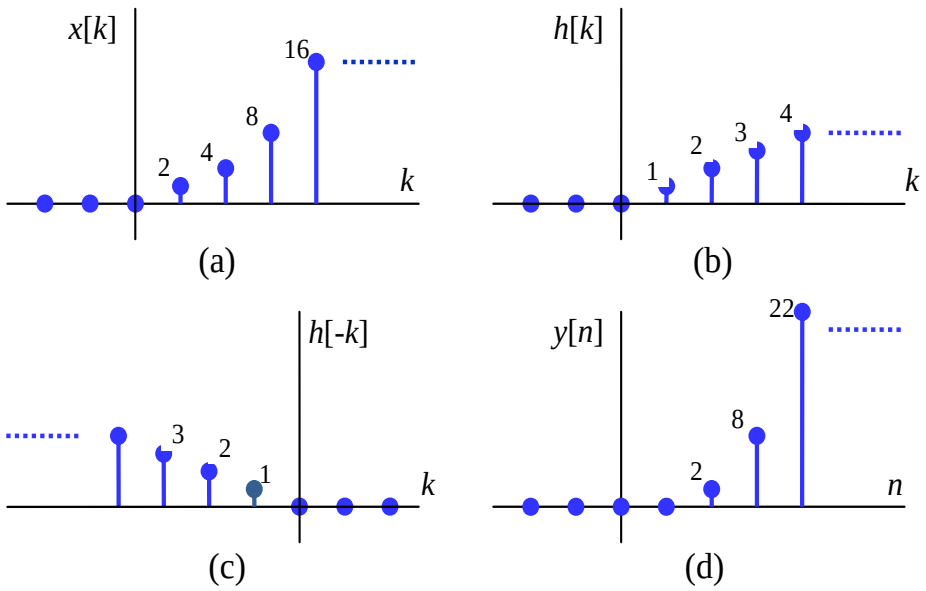

Fig.2.2.4. Example 2.2.2.

As mentioned earlier, a very important property of the convolution operation is its commutative property

$$x[n] * h[n] = h[n] * x[n] \tag{2.2.7}$$

This fact can be proved very easily from the defining equation (2.2.4) by a simple change of variable.

The convolution operation is also associative,

$$x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n] \tag{2.2.8a}$$

and distributive,

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n] \tag{2.2.8b}$$

as can be proved in a straightforward manner.

We already know that the input-output relationship for a LTI system is fully characterized by its impulse response. Because of the associative property of the convolution the cascade or series combination of two LTI systems can be replaced by a single LTI system with the impulse response.

$$h[n] = h_1[n] * h_2[n] \tag{2.2.9}$$

as shown in Figure 2.2.5(a) and (b). Again, using commutative and associative properties one can show that the cascade connection of two LTI systems is also commutative as shown in Fig.2.2.5(c) and (d). Note that for nonlinear or time varying systems this property does not hold in general.

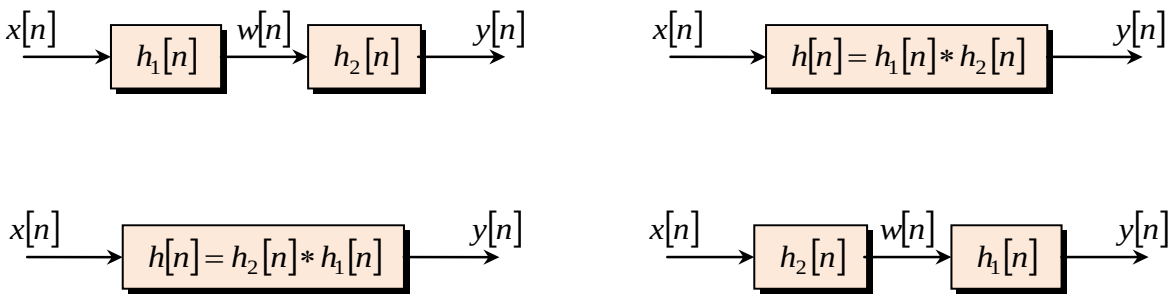

Fig.2.2.5. Associative property of convolution and its implications

Similarly, the distributive property can be used to replace a parallel interconnection of two LTI systems by a single LTI system with the impulse response equal to the summation of individual impulse responses;

$$h[n] = h_1[n] + h_2[n] \tag{2.2.10}$$

as depicted in Figure 2.2.6.

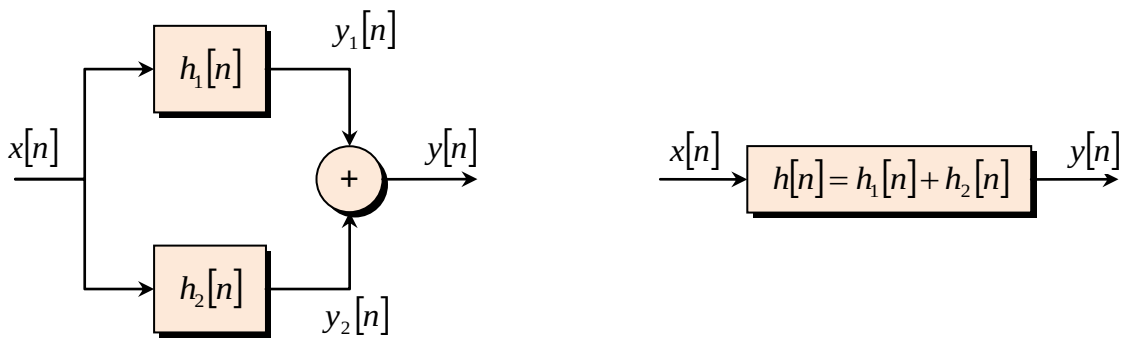

Fig.2.2.6. Interpretation of distributive property for parallel interconnection

Before closing this discussion, it is worth mentioning once again that the results obtained above are valid for LTI systems. While the impulse response completely

characterizes a LTI system, there may be several non-linear systems with the same unit impulse response. Furthermore, while the order in which two LTI systems are cascaded is not important, for nonlinear or time varying systems this property may not hold.

2.3. CONTINUOUS TIME LTI SYSTEMS: THE CONVOLUTION INTEGRAL

The goal of this section is to find a complete characterization for a LTI continuous-time system in terms of its impulse response $h(t)$, in analogy with the discrete-time case.

First, consider the unit-area pulse $\delta_\Delta(t)$ of width Δ depicted in Figure 2.3.l(a), and let the response $h_\Delta(t)$ of a LTI system to $\delta_\Delta(t)$ be as shown in Figure 2.3.l(b). Since the system is linear and time invariant, the response to a pulse scaled and shifted is $h_\Delta(t)$ scaled and shifted by the same amount,

$$x(k\Delta)\delta_\Delta(t-k\Delta) \rightarrow x(k\Delta)h_\Delta(t-k\Delta) \tag{2.3.1}$$

as shown in Figure 2.3.l(c-d).

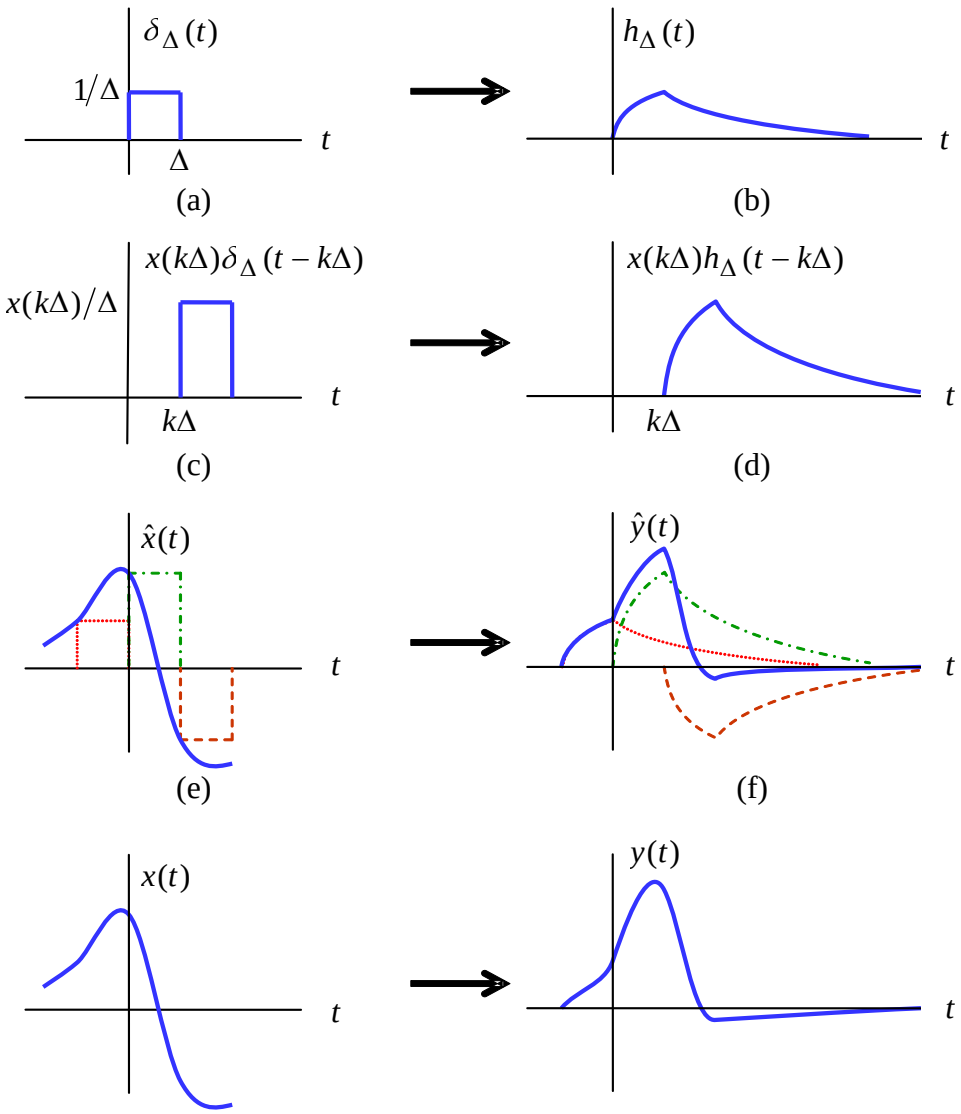

Fig. 2.3.1. Graphical interpretation of the response of an LTI system

In section 2.1 we have seen that a continuous time signal can be approximated by

$$\hat{x}[t] = \sum_{k=-\infty}^{\infty} x(k\Delta)\delta_{\Delta}(t-k\Delta).\Delta \ . \qquad (2.3.2)$$

and as $\Delta \to 0$, $\hat{x}(t) \to x(t)$. Using the additivity property one obtains the response to $\hat{x}(t)$ as

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} x(k\Delta)\delta_{\Delta}(t-k\Delta).\Delta \qquad (2.3.3)$$

as depicted in Figure 2.3.1(e-f). At the limit as $\Delta \to 0$, $\hat{y}(t)$ will approach the actual response $y(t)$ to $x(t)$, as shown in Figure 2.3.1(g-h). The summation on the right hand side of (2.3.3) turns into an integral as $\Delta \to 0$. Thus, the response $y(t)$ can be expressed as

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \ . \qquad (2.3.4)$$

This equation is referred to as the convolution integral or the superposition integral. The convolution of two signals is shown in shorthand as

$$y(t) = x(t) * h(t) \qquad (2.3.5)$$

as in the discrete-time case.

The continuous-time convolution enjoys same properties as its discrete-time counterpart: It is commutative, associative and distributive; i.e.

$$x(t) * h(t) = h(t) * x(t) \qquad \text{(Commutativity) (2.3.6)}$$

$$x(t) * \{h_1(t) * h_2(t)\} = \{x(t) * h_1(t)\} * h_2(t) \qquad \text{(Associativity) (2.3.7)}$$

$$x(t) * \{h_1(t) + h_2(t)\} = x(t) * h_1(t) + x(t) * h_2(t) \qquad \text{(Distributivity). (2.3.8)}$$

As can be seen easily, in continuous-time too, the impulse response completely characterizes a LTI system. Implications of the above properties on the interconnection of LTI systems are the same as the discrete-time case.

The procedure for the calculation of the convolution integral is quite similar to that for the convolution sum. As indicated by (2.3.4), the response *y(t)* for any value of t is a "weighted" integral of the input, where the weight on *x(t)* is *h(t-τ)*. Therefore, to evaluate this integral for a specific value of *t*, we must consider *h(t-τ)* as a function of τ with *t* fixed. This can be done by reflecting or folding $h(\tau)$ about the vertical axis and shifting to the right by *t* (if *t* is positive). We then multiply *x(τ)* and *h(t-τ)* and integrate over $-\infty < \tau < \infty$. Following examples illustrate this procedure.

Example 2.3.1.

Let

$x(t) = e^{-\alpha t}u(t)$

be the input to a LTI system with unit impulse response

$h(t) = u(t)$,

where $\boldsymbol{a}$ *is positive. In Figure 2.3.2.* $h(\tau)$, $x(\tau)$ *and* $h(t-\tau)$ *for t<0 and t>0 are shown. From the figure we see that for t<0, the product* $x(\tau)h(t-\tau)$ *is identically zero and*

therefore $y(t)=0$. For $t>0$ on the other hand, the product $x(\tau)h(t-\tau)$ is as shown in Figure 2.3.2 (e). The area under this function is

$$x(t) = \int_0^t e^{-\alpha\tau} d\tau = -\frac{1}{a} e^{-\alpha\tau}\Big|_0^t = \frac{1}{a}(1 - e^{-\alpha t}).$$

Thus, for all t,

$$y(t) = \frac{1}{a}(1 - e^{-\alpha t})$$

as shown in Figure 2.3.2(f).

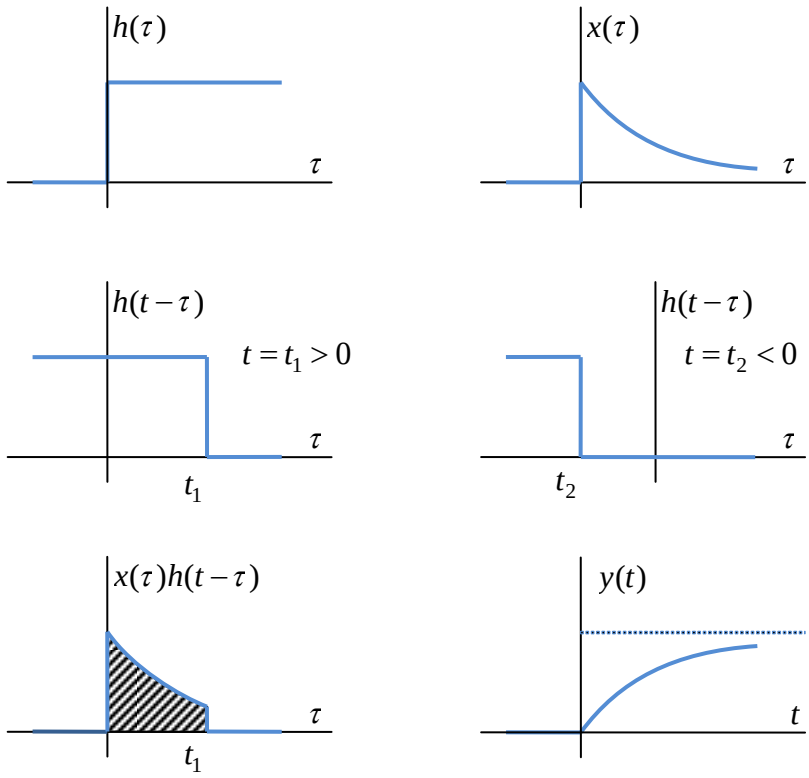

Fig.2.3.2. Example 2.3.1

Example 2.3.2

Let us determine the convolution between

$$x(t) = \begin{cases} 1 & 0 < t < T \\ 0 & otherwise \end{cases}$$

and

$$h(t) = \begin{cases} 1 & 0 < t < 2T \\ 0 & \text{otherwise} \end{cases}.$$

ssssIn Figure 2.3.3 $x(\tau)$and $h(t-\tau)$ for different values of t are shown. In each case the shaded area indicate the area under the product $x(\tau)h(t-\tau)$. Therefore, the convolution $y(\tau)$can be found in 5 different ranges for t as;

$$y(t)=\begin{cases} 0 & t<0 \\ \int_0^t (t-\tau)d\tau=\frac{t^2}{2} & 0<t<T \\ \int_0^T (t-\tau)d\tau=tT-\frac{T^2}{2} & T<t<2T \\ \int_{t-2T}^{T} (t-\tau)d\tau=-\frac{t^2}{2}+tT+\frac{3T^2}{2} & 2T<t<3T \\ 0 & 3T<t \end{cases}.$$

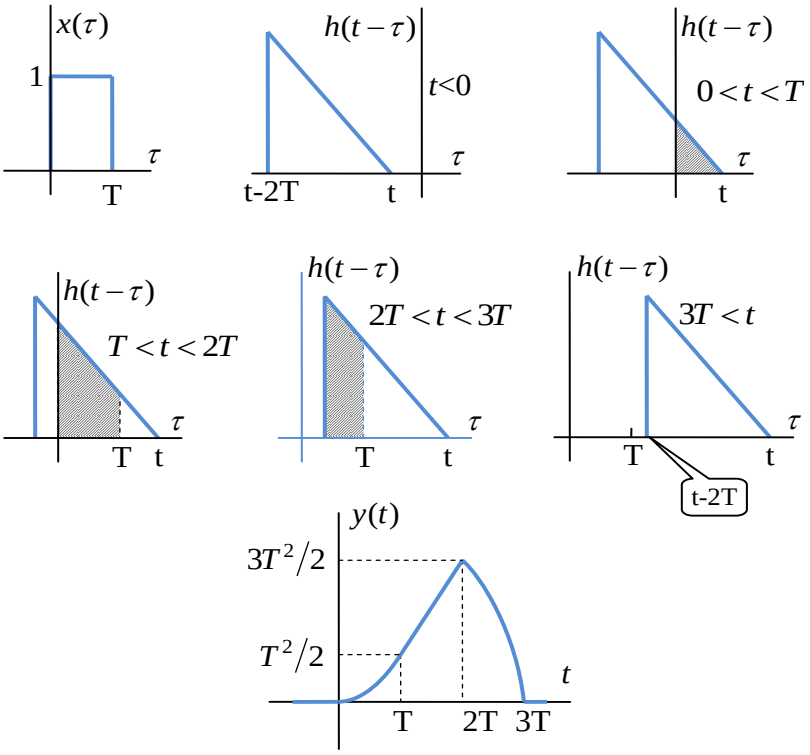

Fig.2.3.3. Example 2.3.2.

Convolution operation has some additional properties when the signals in question satisfy certain special conditions.

In the case where both signals that are being convolved are causal, that is

$$x(t) = x(t)u(t) \tag{2.3.5a}$$

and

$$h(t) = h(t)u(t) \tag{2.3.5b}$$

the formula for convolution becomes

$$y(t) = x(t) * h(t) = \int_0^t x(\tau)h(t-\tau)d\tau. \tag{2.3.6}$$

A more general case which includes this case is when both *x(t)* and *h(t)* vanish at minus infinity, i.e.

$$\lim_{t\to-\infty} x(t) = \lim_{t\to-\infty} h(t) = 0 \ . \tag{2.3.7}$$

For *x(t)* and *h(t)* satisfying (2.3.7), define $\hat{x}(t)$ and $\hat{h}(t)$ as

$$\hat{x}(t) = \int_{-\infty}^{t} x(\tau)d\tau \tag{2.3.8}$$

and

$$\hat{h}(t) = \frac{d}{dt}h(t). \tag{2.3.9}$$

The convolution between $\hat{x}(t)$ and $\hat{h}(t)$ can be found using integration by parts as

$$\begin{aligned}\hat{x}(t) * \hat{h}(t) &= \int_{-\infty}^{\infty}\left\{\int_{-\infty}^{\tau} x(\mu)d\mu\right\}\left\{-\frac{d}{d\tau}h(t-\tau)d\tau\right\} \\ &= \left|-\left\{\int_{-\infty}^{\tau} x(\mu)d\mu\right\}h(t-\tau)\right|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \end{aligned} \tag{2.3.10}$$

where the first term vanishes because of (2.3.7). Therefore for $x(t)$ and $h(t)$ satisfying (2.3.7),

$$x(t) * h(t) = \left\{\int_{-\infty}^{t} x(\mu)d\mu\right\} * \left\{\frac{d}{dt}h(t)\right\}. \tag{2.3.11}$$

The following example illustrates the use of this property.

Example 2.3.3.

Let us evaluate the convolution of Example 2.3.2 using the property (2.3.11). Figure 2.3.4(a) and (b) depict $\int_{-\infty}^{t} h(\mu)d\mu$ *and* $\frac{dx(t)}{dt}$ *respectively. The fact that* $\frac{dx(t)}{dt}$ *is a combination of two impulse functions makes the procedure rather easy:*

$$\begin{aligned} y(t) &= \{\int_{-\infty}^{\infty} x(\tau)d\tau\} * \{\delta(t) - \delta(t-\tau)\} \\ &= \underbrace{\{\int_{-\infty}^{\infty} x(\tau)d\tau\} * \delta(t)}_{y_1(t)} - \underbrace{\{\int_{-\infty}^{\infty} x(\tau)d\tau\} * \delta(t-T)}_{y_2(t)=y_1(t-T)} . \end{aligned}$$

Thus, the result shown in Figure 2.3.4(d) can be obtained from $y_1(t)$ *and* $y_2(t)$ *as shown in Figure 2.3.4(c).*

2.4. PROPERTIES OF LTI SYSTEMS

As we have seen both in discrete-time and in continuous time, the unit impulse response completely characterizes a LTI system. In this section we will examine several important properties of LTI systems in terms of the impulse response.

2.4.1. Memory in LTI Systems

A memoryless system was defined as one, the output of which depends only on the value of the input at the same time instant. In the case of LTI systems, that calls for an impulse response, which is zero everywhere except the origin, i.e.

$$h[n] = K\delta[n] \tag{2.4.1}$$

in discrete-time, or

$$h(t) = K\delta(t) \tag{2.4.2}$$

in continuous-time. Note that in such a case the input-output relationship becomes

$$y[n] = Kx[n] \tag{2.43}$$

or

$$y(t) = Kx(t) \tag{2.4.4}$$

depending on whether one is dealing with a discrete-time or continuous-time system.

In the special case K=l, the system is what's-so-called an "identity system", i.e. a

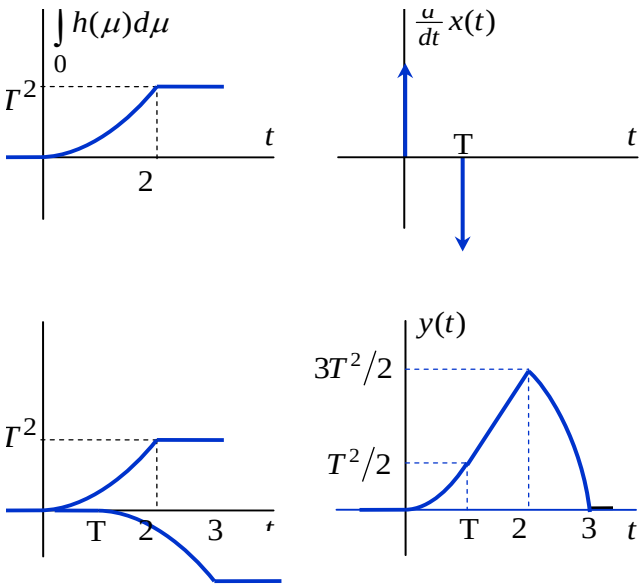

Fig.2.3.4. Example 2.3.3.

system for which the input and the output are always identical.

2.4.2. Causality in LTI Systems

A causal system was defined as one the output of which depends only on the past and present values of the input. In the case of a LTI system, the output is related to the input by convolution. Considering the discrete-time case, when the input is a "causal signal", i.e.

$$x[n] = x[n]u[n] \tag{2.4.5}$$

the output is

$$y[n] = \sum_{k=-\infty}^{n} x[n-k]h[k] \ . \tag{2.4.6}$$

This output will be zero for $n<0$ for any causal $x[n]$ if and only if

$$h[n] = 0 \quad \text{for } n<0 \ . \tag{2.4.7}$$

Therefore, a discrete-time LTI system is causal if and only if (2.4.7) holds. Similarly, the necessary and sufficient condition for the causality of a LTI continuous-time system is

$$h(t) = 0 \quad \text{for } t<0. \tag{2.4.8}$$

2.4.3. Stability of LTI Systems

As discussed earlier, a system is stable if every bounded input yields a bounded output. To determine the conditions of stability for LTI systems, let us consider a

$$|x[n]| \le B < \infty . \tag{2.4.9}$$

In this case, the magnitude of the output satisfies

$$\begin{aligned} |y[n]| &= \left| \sum_{k=-\infty}^{\infty} h[k]x[n-k] \right| \\ &\le \sum_{-\infty}^{\infty} |h[k]| . |x[n-k]| \\ &\le B \sum_{k=-\infty}^{\infty} |h[k]| \end{aligned} \tag{2.4.10}$$

Obviously, $y[n]$ is bounded if the unit impulse response is "absolutely summable", i.e. if

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty . \tag{2.4.11}$$

Here we have only shown that (2.4.11) is a sufficient condition for the stability of a LTI system. In fact, this condition is also necessary, for in the case it is not satisfied, one can find bounded inputs resulting in unbounded outputs.

An analogous condition is valid for continuous-time LTI systems. A continuous-time LTI system is stable if and only if its impulse response is "absolutely integrable", i.e.

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau . \tag{2.4.12}$$

Example 2.4.1

Consider the pure time shift in discrete time:

$y[n] = x[n-1]$

which has the unit impulse response

$h[n] = \delta[n-1]$.

Since

$$\sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=-\infty}^{\infty} |\delta[n-1]| = 1 < \infty$$

is bounded, the pure time shift is a stable system.

Example 2.4.2.

A continuous-time "integrator" is defined by the relationship

$$y(t) = \int_{-\infty}^{t} x(\tau) d\tau \ .$$

The impulse response of this system;

$$h(t) = \int_{-\infty}^{t} \delta(\tau) d\tau = u(t)$$

is not absolutely integrable since

$$\int_{-\infty}^{\infty} |u(\tau)|\, d\tau = \int_{0}^{\infty} d\tau = \infty \ ,$$

and therefore the "integrator" is not a stable system.

2.4.4. The Unit Step Response

We have seen that, the unit impulse response completely characterizes a LT1 system. It is possible to characterize properties like causality, stability etc. in terms of the impulse response.

There is another input signal, that is used quite often to characterize a LTI system. This is the unit step response $g[n]$ or $g(t)$

Since

$$u[n] = \sum_{k=-\infty}^{n} \delta[k] \ , \tag{2.4.13}$$

using the superposition property, one obtains the step response as;

$$g[n] = \sum_{k=-\infty}^{n} h[k] \ . \tag{2.4.14}$$

Then it is a very simple task to show that

$$h[n] = g[n] - g[n-1] \tag{2.4.15}$$

Thus, the unit step response completely characterizes the unit impulse response and therefore the corresponding LTI system.

Similarly, the unit step response $g(t)$ and unit impulse response $h(t)$ of a continuous-time LTI system are related by

$$g(t) = \int_{-\infty}^{t} h(\tau) d\tau \tag{2.4.16}$$

and

$$h(t) = \frac{dg(t)}{dt} = g'(t) \ . \tag{2.4.17}$$

And thus, the step response fully characterizes the LTI system.

2.5. SYSTEMS DESCRIBED BY DIFFERENTIAL AND DIFFERENCE EQUATIONS

In a most general and simple minded way, the input-output relationship of a system can be described by an exhaustive listing of all possible input signals and their corresponding outputs. However, an exhaustive listing is impossible to generate. Therefore, we usually try to express this relationship in a compact way. If the input-output relationship can be described by an equation or a set of equations, for a given input signal, the output can be determined by solving this equation. For example, assume that the input sequence $x[n]$ and the output sequence $y[n]$ of a discrete-time system are related by the equation

$$y[n] = x[n+1] + ax[n] + bx[n-1] \ . \tag{2.5.1}$$

Then for a given $x[n]$, $y[n]$ can be calculated for all $\boldsymbol{n}$. Similarly the response of a continuous-time system described by

$$y(t) = x(t) + a.t.\frac{dx(t)}{dt} \tag{2.5.2}$$

to an input signal $x(t)$ can be found by evaluating this expression for $y(t)$.

These two examples are rather trivial, since in both cases the output is already explicitly expressed in terms of the input. In a more general case, the output signal and the input signal may be related in an implicit way, such as

$$\frac{dy(t)}{dt} + ay(t) = tx(t) \tag{2.5.3}$$

or

$$y^2[n] + ny[n-1] = x[n] + x[n-1]\ . \tag{2.5.4}$$

In this section, we will investigate input-output relationships describable in the form of a differential or difference equation, the examples for which are given in (2.5.3) and (2.5.4) respectively. For a continuous-time system, a differential equation description relates the values of the input signal $x(t)$ and a number of its derivatives to the values of the output signal $y(t)$ and its derivatives, for all $\boldsymbol{t}$. Similarly for a discrete-time system a difference equation relates the successive values of the input, $x[n]$, $x[n-1]$,… to those of the output $y[n]$, $y[n-1]$,... for all $\boldsymbol{n}$. The output for a given input can be obtained by solving the related differential or difference equation for the output variable $y(t)$ ($y[n]$) with $x(t)$ ($x[n]$) replaced by the given input.

Unfortunately, there is no general method for the solution of a general differential or difference equation. Here, we will restrict out attention to linear, constant coefficient differential or difference equations.

2.5.1. Linear Differential Equations with Constant Coefficients

Let us start our discussion with a simple example. Consider a continuous-time system with input-output relationship

$$\frac{dy(t)}{dt} + ay(t) = x(t)\ , \tag{2.5.5}$$

where $\boldsymbol{a}$ is a real number. Here the response is expressed implicitly. One must solve the differential equation to obtain an explicit expression for $y(t)$, in terms of the input $x(t)$. Let us try to find the response to

$$x(t) = Ke^{-bt}u(t) \tag{2.5.6}$$

where $\boldsymbol{K}$ and $b \neq a$ are real numbers.

The complete solution of this equation consists of two parts: the "particular solution", $y_p(t)$ and the "homogenous solution", $y_h(t)$; that is

$$y(t) = y_p(t) + y_h(t)\ . \tag{2.5.7}$$

Here $y_h(t)$ is a solution of the homogenous equation

$$\frac{dy_h(t)}{dt} + ay_h(t) = 0 \tag{2.5.8}$$

and $y_p(t)$ satisfies (2.5.5). For the problem at hand, the assumption

$$y_p(t) = Ce^{-bt}, \tag{2.5.9}$$

when substituted in (2.5.5), yields

$$-Cbe^{-bt} + Cae^{-bt} = Ke^{-bt} \tag{2.5.10}$$

and thus

$$C = \frac{K}{a-b} \quad . \tag{2.5.11}$$

Therefore the particular solution is

$$y_p(t) = \frac{K}{a-b}.e^{-bt}u(t) \ . \tag{2.5.12}$$

In order to determine $y_h(t)$, we assume a solution of the form

$$y_h(t) = Ae^{st} \ . \tag{2.5.13}$$

The substitution of the assumed solution in (2.5.8) yields

$$Ase^{st} + aAe^{st} = 0 \tag{2.5.14}$$

which is satisfied for s=-a. Therefore the homogenous solution is

$$y_h(t) = Ae^{-at}. \tag{2.5.15}$$

Combining (2.5.12) and (2.5.15) as in (2.5.7) one obtains the general solution as

$$y(t) = Ae^{-at} + \frac{K}{a-b}.e^{-bt}. \tag{2.5.16}$$

Note that this solution does not completely specify $y(t)$in terms of the input $x(t)$; the coefficient ***A*** has not been determined yet. For the input to specify the output completely, some auxiliary conditions on the differential equation must be specified. For the first order differential equation in our example specification of $y(t)$alone for a given instant of time will suffice to specify $y(t)$completely for all t. However, note that in this case, $y(t)$will be equal to the homogenous solution for t<0, since $x(t)$ is a causal signal. The reader is encouraged to show that the continuous-time system thus defined is neither causal, nor linear although (2.5.5) is a linear differential equation.

It is possible to impose an additional auxiliary condition in a manner such that he resulting continuous-time system is causal and linear. That is the condition of "initial rest". We will say that a system is initially at rest if $y(t)$ is zero for $t < t_0$, for any $x(t)$ which is zero for $t < t_0$. It is left to the reader to show that a system defined by a differential equation and the initial rest assumption is both linear and causal.

Obviously, if we make the initial rest assumption and if $x(t) = 0$ for $t < t_0$, then we have to solve the differential equation for $t < t_0$ only, for

$$y(t) = 0 \tag{2.5.17}$$

which is called the "initial condition".

In addition to causality and linearity, the initial rest condition also guarantees

the time-invariance of a system described by a constant coefficient differential equation. To illustrate this consider the system described by (2.5.2) together with the initial rest condition. Let $y_1(t)$ be the response to $x_1(t)$ which is zero for $t < t_0$, i.e.

$$\frac{d}{dt}y_1(t) + ay_1(t) = x_1(t) \tag{2.5.18a}$$

and

$$y_1(t) = 0 \quad t \leq t_0 \ . \tag{2.5.18b}$$

Now consider the shifted input

$$x_2(t) = x_1(t - T). \tag{2.5.19}$$

Clearly $x_2(t) = 0$ for $t \leq t_0 + T$. For the system to be time-invariant, the solution to

$$\frac{d}{dt} y_2(t) + a y_2(t) = x_2(t) \tag{2.5.20a}$$

with

$$y_2(t) = 0 \quad for \quad t \leq t_0 \tag{2.5.20b}$$

must be the shifted version of $y_1(t)$, i.e.

$$y_2(t) = y_1(t - T)\ . \tag{2.5.21}$$

The straightforward proof of this assertion is left to the reader as an exercise.

The general N'th order linear, constant coefficient differential equation

$$\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^{N} b_k \frac{d^k}{dt^k} x(t) \tag{2.5.22}$$

together with a set of N auxiliary conditions corresponding to the values of

$$y(t),\ \frac{d}{dx} y(t), \ldots\ldots\ldots \frac{d^{N-1}}{dx^{N-1}} y(t) \tag{2.5.23}$$

at some point in time will specify a system completely, A system thus specified will be causal, linear and time invariant if we assume initial rest, i.e. if

$$y(t_0) = \frac{d}{dx} y(t_0) = \ldots\ldots\ldots = \frac{d^{N-1}}{dx^{N-1}} y(t_0) = 0 \tag{2.5.24}$$

for any $x(t)$ which is zero for $t \leq t_0$.

From now on, we will focus our attention on LTI systems and therefore whenever the system is described by a differential equation, the initial rest assumption will be made.

2.5.2. Linear Difference Equations with Constant Coefficients

A general N'th order linear, constant coefficient difference equation is given as

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k] \tag{2.5.25}$$

Such an equation can be solved in a manner exactly analogous to the solution of a differential equation. Specifically, the solution $y[n]$ consists of "particular solution" $y_p[n]$ satisfying (2.5.25) and the "homogenous solution" $y_h[n]$ satisfying the "homogenous equation"

$$\sum_{k=0}^{N} a_k y[n-k] = 0\ . \tag{2.5.26}$$

The method of solution is illustrated by the following example:

Example 2.5.1

Let us determine the general solution of the second order difference equation

$$y[n] - \frac{5}{6} y[n-1] + \frac{1}{6} y[n-2] = x[n]$$

for

$x[n] = 3^n u[n]$.

The homogenous solution satisfies

$y_h[n] - \frac{5}{6} y_h[n-1] + \frac{1}{6} y_h[n-2] = 0$.

A solution of the form

$y_h[n] = A.r^n$

satisfies

$$A.r^n - \frac{5}{6} A.r^{n-1} + \frac{1}{6} A.r^{n-2} = A.r^{n-2} \frac{1}{6}\{6r^2 - 5r + 1\}$$

For $r = 1/2$ *and* $r = 1/3$, *and sufficiently large n. Thus the homogeneous solution is*

$$y_h[n] = c_1 \left(\frac{1}{2}\right)^{\mathrm{n}} + c_2 \left(\frac{1}{3}\right)^{\mathrm{n}}.$$

On the other hand

$y_h[n] = b.3^n u[n]$

is a particular solution if

$b.3^n - \frac{5}{6} b.3^{n-1} + \frac{1}{6} b.3^{n-2} = 3^n$,

that is when

$b = \frac{27}{20}$.

Therefore the general solution is

$$y[n] = y_h[n] + y_p[n] = c_1 \left(\frac{1}{2}\right)^{\mathrm{n}} + c_2 \left(\frac{1}{3}\right)^{\mathrm{n}} + \frac{27}{20} 3^n u[n] \ .$$

Although a difference equation can be solved in a manner analogous to differential equations, as the example above illustrates, there is yet another method for the solution of difference equations. (2.5.21) can be rearranged as

$$y[n] = \frac{1}{a_0}\left\{\sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k]\right\} \qquad (2.5.27)$$

This equation gives the "present" value of the solution at time n in terms of the past values of $y[n]$ and the present and past values of $x[n]$. This way the solution can be determined step by step. The necessity for auxiliary conditions is obvious here. In order to determine $y[n]$, the knowledge of $y[n]$, $y[n-1]$, $y[n-2]$,..., $y[n-N]$ is necessary.

A difference equation together with auxiliary conditions defines a system. In the case the input is zero, the output is zero only if the auxiliary conditions are zero. So, although the difference equation is linear, the system it defines will be nonlinear for nonzero auxiliary conditions. In order to have a linear system we will again need the initial rest assumption. A system is said to be initially at rest if the response $y[n]$ to an input $x[n]$ which is zero for $n \leq n_0$, is zero for $n \leq n_0$. It can be shown in the usual manner that the initial rest assumption guarantees time invariance and causality in addition to linearity.

2.6. DETERMINATION OF THE IMPULSE RESPONSE FROM THE DIFFERENTIAI OR DIFFERENCE EQUATION

When a LTI system is described by a difference or differential equation and initial rest assumption, it is always possible to find the response to a given input by solving the equation by classical techniques mentioned in the previous chapter. Another technique, which is more powerful, is the determination of the impulse response. Once the impulse response is known, one can determine the response to any input, simply by convolving it with the impulse response.

The impulse response can be obtained directly by solving the given difference or differential equation for the unit impulse as the input signal. The solution of such an

equation is somewhat simpler than the general case since it basically involves the homogenous solution of the equation. Once the equation is solved for an impulse input, the response to differing inputs can be determined directly by convolution, with no need to solve the equation again.

2.6.1. Finding the Impulse Response Function from the Differential Equation

To develop the method, suppose that we have a second order system:

$$y''(t) + a_1 y'(t) + a_0 y(t) = x(t) \tag{2.6.1}$$

where prime indicates differentiation with respect to ***t***.

The impulse response will satisfy the equation

$$h''(t) + a_1 h'(t) + a_0 h(t) = \delta(t)\ . \tag{2.6.2}$$

Since the unit impulse function is zero everywhere except at the origin, satisfies the homogenous equation

$$h''(t) + a_1 h'(t) + a_0 h(t) = 0 \tag{2.6.3}$$

for ***t*** strictly positive. The general solution for (2.6.3) can be found in the usual manner. The coefficients in the general solution must be determined from the initial conditions, which are in turn, fixed by the initial rest assumption.

Returning to (2.6.3), two linearly independent solutions can be obtained from the characteristic equation and the homogenous solution is a linear combination of these two solutions. Remembering that, initial rest assumption guarantees the causality of a system described by a linear, constant coefficient differential equation, for $t < 0$. Therefore, the solution of (2.6.3) has the form

$$h(t) = \{c_1 e^{s_1 t} + c_2 e^{s_2 t}\} u(t)\ . \tag{2.6.4}$$

Undetermined coefficients c_1 and c_2 are to be selected such that (2.6.2) is satisfied at $t = 0$ also, the only point in time, that we have not considered so far.

A closer inspection of (2.6.2) reveals that none of the terms on the left hand side contain the derivatives of the impulse function. Furthermore only the highest derivative of $h(t)$ contains an impulse and other terms are bounded for small values of ***t***. Therefore, integrating (2.6.2) from 0^- to 0^+ one obtains

$$\int_{0^-}^{0^+} h''(\tau) d\tau + a_1 \int_{0^-}^{0^+} h'(\tau)\, d\tau + a_0 \int_{0^-}^{0^+} h(\tau) d\tau = 1\ . \tag{2.6.5}$$

Second and third integrals are zero because of boundedness of the integrand. The first integral on the other hand is equal to $h'(0^+)$, and one has

$$h'(0^+) = 1 \tag{2.6.6}$$

since $h'(0^-) = 0$ by initial rest assumption. On the other hand,

$$h(0^+) = \int_{0^-}^{0^+} h'(\tau)\, d\tau$$

since $h'(\tau)$ is bounded for small t.

As a result, we conclude that for $t \geq 0$ the unit impulse response of (2.6.1) is the solution of the homogenous equation (2.6.3) with the initial conditions

$$h(0) = 0, \qquad h'(0) = 1 \tag{2.6.8}$$

and it is equal to zero for $t < 0$.

Example 2.6.1

Consider the LTI system described by

$$y''(t) + 7y'(t) + 12y(t) = e^t u(t).$$

Let us find the impulse response first. The impulse response $h(t)$ satisfies

$$h''(t) + 7h'(t) + 12h(t) = 0$$

with $h(0) = 0, \qquad h'(0) = 1.$

The characteristic equation is

$$s^2 + 7s + 12 = 0$$

which is satisfied for s=-3 and s=-4. Therefore, one has

$$h(t) = \begin{cases} c_1 e^{-3t} + c_2 e^{-4t} & \text{for } t \geq 0 \\ 0 & \text{for } t < 0 \end{cases}.$$

When the initial conditions are used

$$h(0) = 0 = c_1 + c_2$$

$$h'(0) = 1 = -3c_1 - 4c_2$$

and the coefficients are determined as

$$c_1 = 1, \quad c_2 = -1.$$

Therefore,

$$h(t) = (e^{-3t} - e^{-4t})u(t).$$

For the input given,

$$\begin{aligned} y(t) &= x(t) * h(t) \\ &= \{(e^{-3t} - e^{-4t})u(t)\} * \{e^t u(t)\} \\ &= \int_0^t (e^{-3\tau} - e^{-4\tau})\, e^{t-\tau} d\tau = e^t \int_0^t (e^{-4\tau} - e^{-5\tau}) d\tau \\ &= e^t \left[-\frac{1}{4} e^{-4t} + \frac{1}{5} e^{-5t} \right]_0^t = \frac{1}{20}(4e^{-4t} - 5e^{-3t} + e^t) \end{aligned}.$$

The idea presented above can readily be extended to a LTI system described by an N'th order differential equation

$$\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y(t) = x(t) \quad . \tag{2.6.9}$$

In this case, we find that $h(t)$ satisfies

$$\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} h(t) = 0$$

with $h(0) = h'(0) = ...h^{(N-2)}(0) = 0$ and $h^{(N-1)}(0) = 1$ for $t \geq 0$ and $h(t) = 0$ for $t < 0$.

Example 2.6.2

Consider

$y'''(t) - 3y''(t) + 3y'(t) - y(t) = x(t)$.

The impulse response satisfies

$h'''(t) - 3h''(t) + 3h'(t) - h(t) = 0$

with $h(0) = h'(0) = 0$ *and* $h''(0) = 1$.

The characteristic equation

$s^3 - 3s^2 + 3s - 1 = (s-1)^3$

has a triple root at s=l. Therefore

$h(t) = c_1 e^t + c_2 t e^t + c_3 t^2 e^t$ *for* $t \geq 0$.

When this solution is matched with initial conditions

$h(0) = 0 = c_1$

$h'^{(0)} = 0 = c_{1+} c_2$

$h''^{(0)} = 1 = c_1 + 2c_2 + 2c_3$

And one gets $c_1 = c_2 = 0$ *and ,* $c_3 = {}^1/_2$. *Thus*

$h(t) = \frac{1}{2} t^2 e^t u(t)$.

2.6.2. Finding the Impulse Response Sequence from the Difference Equation

The impulse response sequence of a LTI discrete-time system described by a difference equation

$$\sum_{k=0}^{N} a_k y[n-k] = x[n]$$

can be determined in a manner analogous to the one given above. Obviously, the impulse response is a causal sequence satisfying

$$\sum_{k=0}^{N} a_k h[n-k] = \delta[n] \quad . \tag{2.6.10}$$

Since $\delta[n] = 0$ for $n > 0$, the impulse response satisfies the homogenous equation for $n > 0$. That is, $h[n]$ can be expressed as a linear combination of N linearly independent sequences each of which satisfy the homogenous equation. However, the coefficients of the linear combination are yet to be determined from the initial conditions on $h[n]$. Because of the initial rest assumption

$$h[-1] = h[-2] = \ldots\ldots\ldots\ldots h[-N+1] = 0.$$

On the other hand since $\delta[0] = 1$, (2.6.10) for $n = 0$ yield $h[0] = 1$. Therefore, the impulse response sequence of a LTI discrete time system is the solution of the homogenous equation with

$$h[-1] = h[-2] = \ldots\ldots\ldots\ldots h[-N+1] = 0$$
$$h[0] = 1$$

for $n \geq 0$ and is zero for $n < 0$.

Example 2.6.3

Let us determine the impulse response of the LTI system described by

$y[n] - \frac{1}{4}y[n-2] = x[n-1]$.

Note here that the right hand side is not $x[n]$ but $x[n-1]$.

If the right hand side was $x[n]$ the impulse response sequence $\hat{h}[n]$ would satisfy

$$\hat{h}[n] - \frac{1}{4}\hat{h}[n-2] = 0$$

with $\hat{h}[0] = 1$, $\hat{h}[-1] = 0$.

The characteristic equation

$$r^2 - \frac{1}{4}$$

has roots $r_{1,2} = \mp {}^{1}/_{2}$ and therefore

$$\hat{h}[n] = c_1\left({}^{1}/_{2}\right)^n + c_2\left(-{}^{1}/_{2}\right)^n$$

for $n > 0$. Initial conditions are also satisfied if

$$\hat{h}[0] = c_1 + c_2 = 1$$

and

$$\hat{h}[-1] = 2c_1 - 2c_2 = 0.$$

Therefore the complete solution for $\hat{h}[n]$ is

$$\hat{h}[n] = \left\{\left(\tfrac{1}{2}\right)^{n+1} - \left(-\tfrac{1}{2}\right)^{n+1}\right\}u[n] = \left(\tfrac{1}{2}\right)^n\{1-(-1)^n\}u[n].$$

Now noting that the right hand side is $x[n-1]$, but not $x[n]$, from the time-invariance of the system

$$h[n] = \hat{h}[n-1] = \left(\frac{1}{2}\right)^{n-1}\{1+(-1)^n\}u[n-1]$$

$$= \begin{cases} 0 & n \leq 0 \\ 0 & n \text{ even} \\ \left({}^{1}/_{2}\right)^{n-1} & n > 0 \text{ and odd} \end{cases}.$$

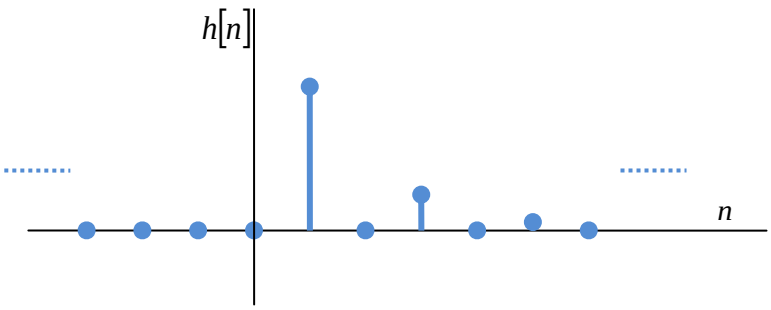

Fig.2.6.1 Example 2.6.3

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(*) of primary importance

PROBLEMS

Problem 1.

Find the general solution of the differential equation

$$\frac{d^4}{dt^4}y(t) + 2\frac{d^2}{dt^2}y(t) + y(t) = 0 \quad .$$

Problem 2.

The general solution of the differential equation

$$\frac{d^3y(t)}{dt^3} + \frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} + y(t) = 0$$

can be written as

$$y(t) = c_1 e^{-t} + c_2 \cos t + c_3 \sin t \quad .$$

Find the impulse response of the system

$$\frac{d^3y(t)}{dt^3} + \frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} + y(t) = \frac{dx(t)}{dt} \quad .$$

Problem 3.

$x[n] = n,\ n \geq 0$ satisfies the difference equation

$$x[n+2] - 2x[n+1] + x[n] = 0 \ .$$

Find the general solution $y[n]$ of the difference equation

$$y[n+2] - 2y[n+1] + y[n] = n, \quad n \geq 0 \ .$$

Identify homogenous and particular solutions in your answer.

Problem 4.

Determine the convolution sum or integral (as appropriate), for the following pairs of functions:

i) $x[n] = a^n.u[n]$, and $h[n] = b^n.u[n]$,

ii) $x[n] = h[n] = a^n.u[n]$,

iii) $x[n] = \left(-\frac{1}{2}\right)^n u[n-2]$, and $h[n] = 2^n u[2-n]$,

iv) $x(t) = Sin(\pi t)\{u(t) - u(t-2)\}$, and $h(t) = u(t) - u(t-1)$,

v) $x(t) = \begin{Bmatrix} e^t & for\ t > 0 \\ 2e^{-t} - e^{2t} & for\ t < 0 \end{Bmatrix}$ and $h(t) = u(t+1)$

vi) $x(t) = u(t) - (t-1)u(t-1) + (t-2)u(t-2)$, and

$h(t) = (t+1)u(t+1) - (t-1)u(t-1) - u(t-2) - 1.$

Problem 5.

Find the impulse response sequence of the discrete-time system described by the difference equation

$$y[n] - 2y[n-1] + y[n-2] = x[n] - x[n-2] \ .$$

Problem 6.

Consider a first-order **LTI** system with an apriori specified unknown initial state. An experiment performed on this system indicates that the response to a unit step is

$$y(t) = \frac{1}{3} + \frac{11}{3}e^{-3t} \qquad for\, t > 0 \ .$$

a) Find the zero-state unit step response.
b) Find the zero-input response for the given initial state.
c) Find the zero-state unit impulse response.

Problem 7.

Consider the LTI system described by the difference equation

$$15y[n] - 8y[n-1] + y[n-2] = 5x[n] + x[n-1]$$

along with the initial rest assumption.

a) Find the impulse response of this system.
b) State whether the system is causal, stable and has memory.

Problem 8.

A linear, time-invariant, causal system is described by the difference equation

$$y[n] + y[n-1] - 2y[n-2] = x[n] - x[n-1].$$

a) Determine an expression for the unit sample (impulse) response of this system.

b) Sketch the unit sample response for $-3 \le n \le 5$.

Problem 9.

A linear, time-invariant causal system is characterized by the differential equation

$$\ddot{y}(t) - \frac{1}{4}y(t) = x(t).$$

a) Determine the impulse response of this system.
b) Is this system stable? Explain.
c) Making use of the above, determine the impulse response of the LTI, causal system described by the differential equation

$$\ddot{y}(t) - \frac{1}{4}y(t) = \dot{x}(t).$$

Problem 10.

Determine and sketch the convolution between the functions $f(t)$ and $g(t)$ given in Figure P10.

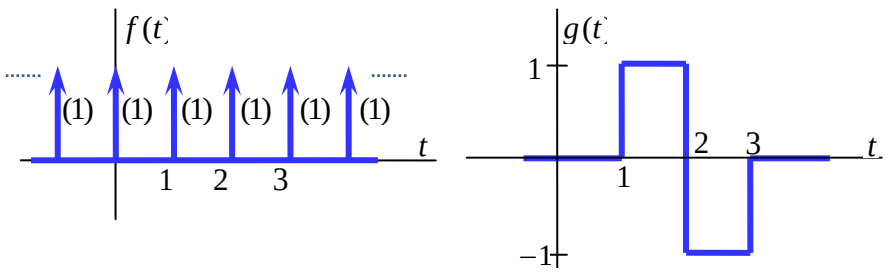


Figure 10

Problem 11.

Determine and sketch the convolution of $f(t)$ given in Figure P11, by itself.

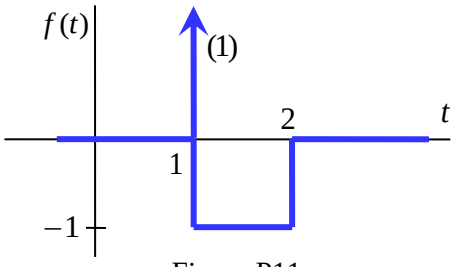


Figure P11

Problem 12.

A linear, time-invariant system is known to have the input-output relationship indicated as:

$$x(t) = e^{j\omega t} \longrightarrow y(t) = \begin{cases} e^{j\omega t} & \text{if } |\omega| \le \omega_0 \\ 0 & \text{if } |\omega| > \omega_0 \end{cases}$$

where ω_0 is a real number.

a) Find the impulse response $h(t)$ of this system,

b) Is this system causal? Prove your answer.

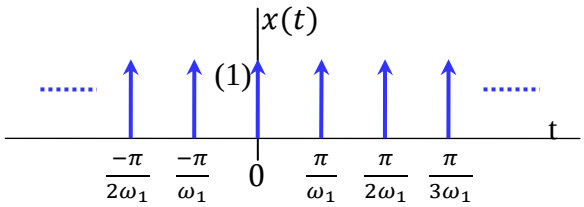


Figure P12

c) Let the periodic impulse train $x(t)$ in the figure be the input to this system. Find the output $y(t)$ (time-domain expression) if $\omega_0 = 10.\omega_1$.

Problem 13.

The impulse response sequence of a given **LTI** discrete-time system is

$$h[n] = (-1)^n u[n].$$

Determine the response of this system to an input sequence

$$h[n] = \begin{cases} 1 & n = 0 \ or 1 \\ -1 & n = 2 \ or 3 \\ 0 & otherwise \end{cases} .$$

Problem 14.

Show that if $x(t)$ and $y(t)$ are periodic signals with periods T_x and T_y respectively, then their convolution $w(t) = x(t) * y(t)$ is also periodic and determine the period of $w(t)$ in terms of periods of $x(t)$ and $y(t)$.

Problem 15.

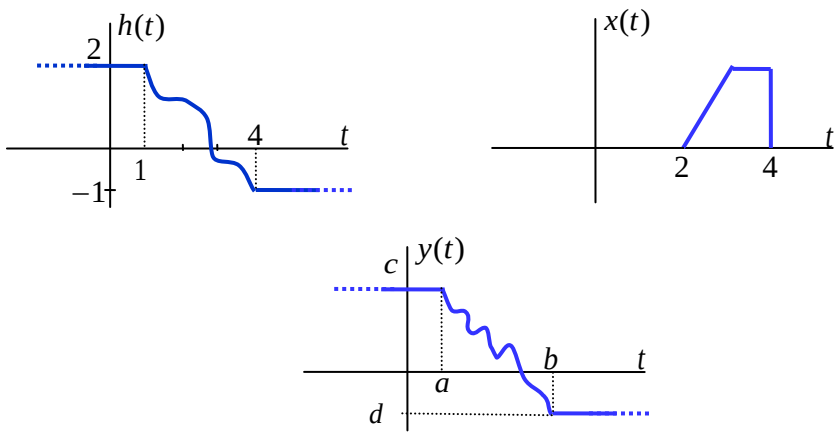


Figure P15

A **LTI** system with the unit impulse response $h(t)$ is excited by an input $x(t)$ as partially shown in Figure P13 (a) and (b) respectively. Determine the parameters **a,b,c** and **d** of the output $y(t)$ shown in Figure P13(c).

Problem 16.

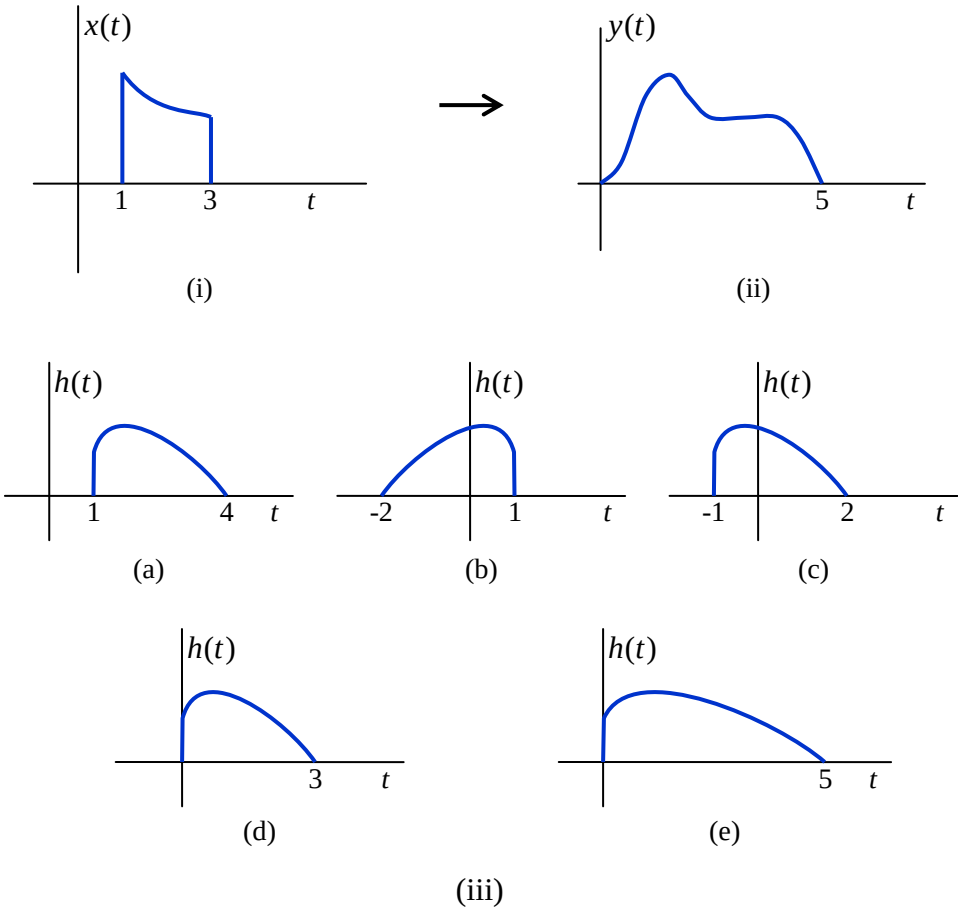
A continuous-time system is described by the input-output relationship

$$x(t) \longrightarrow y(t) = x(t) * x(t)$$

Determine (give a counter example if your answer is negative, prove if positive) whether this system is

- **i)** Memoryless,
- **ii)** Causal,
- **iii)** Linear,
- **iv)** Time-invariant,
- **v)** Stable.

Problem 17.



A **LTI** system is excited by $x(t)$ to yield an output $y(t) = x(t) * h(t)$ as shown in Figure 17 (i) and (ii). Determine which one of the five unit impulse responses shown in Figure P17 (iii)(a-e) could possibly characterize this system.

Problem 18.

Consider the discrete-time system described by the input-output relationship below.

($x[n]$: input; $y[n]$: output):

$$y[n] = \begin{cases} x[n] & if\ x[n+1] \geq 0 \\ x[n-1] & if\ x[n+1] < 0 \end{cases}$$

i) Is this system linear? Prove your answer.
ii) Is it time-invariant? Prove your answer,
iii) Is it causal? Prove your answer.
iv) Is it stable? Prove your answer,
v) Let $h[n]$ be the impulse response of this system. Can the output be expressed as a convolution sum of the input with $h[n]$? Explain why.

Problem 19.

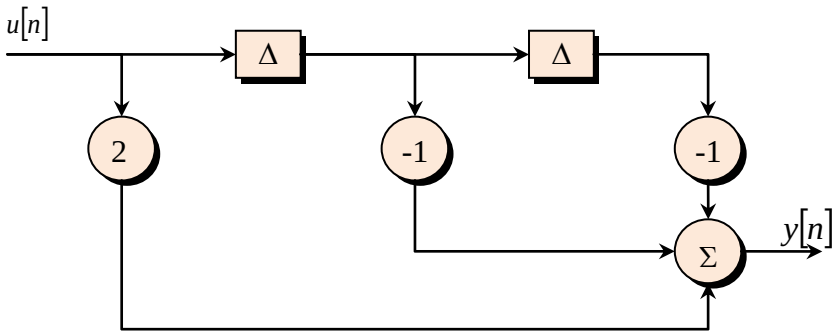


Figure P18

The impulse response sequence of the discrete-time system shown in Figure P18 is

$\{h[n]\} = \{\underline{2}, 1, 1, 0, 0, 0, \cdots\}$. (Note: Under-bar indicates the value for n=0.)

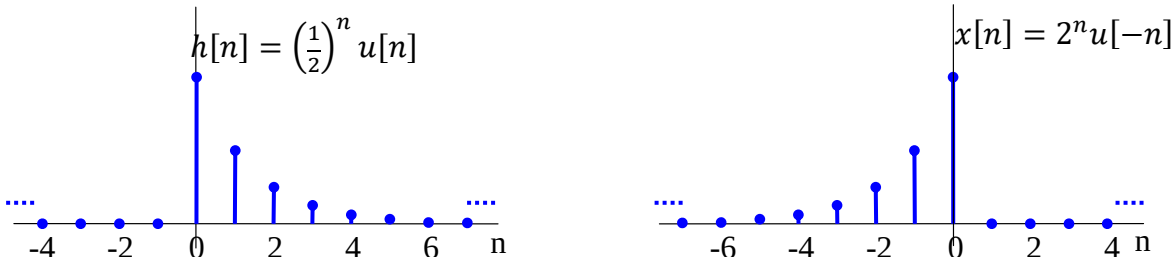
Find the output sequence $\{y[n]\}$ corresponding to an input sequence given by

$$u[n] = \begin{cases} \frac{1}{2^n} & n \geq 0 \\ 0 & n < 0 \end{cases} .$$

Give first five terms of the sequence explicitly.

Problem 20.

A LTI system is characterized by the impulse response $h[n]$shown in figure (a). Determine and sketch its response to the input $x[n]$ shown in figure (b).



(a) (b)

Figure P20

a) Is this system stable?

b) Is it causal?

Explain.

Problem 21.

Find the impulse responses $h_1[n]$ and $h_2[n]$ of the causal and LTI systems S1 and S2. Also

$$S1: \quad y[n] = \sum_{k=-\infty}^{n} x[k] \qquad\qquad S2: \quad z[n+1] + 2z[n] = y[n-1]$$

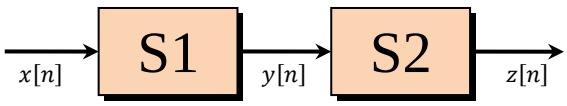


Figure P21

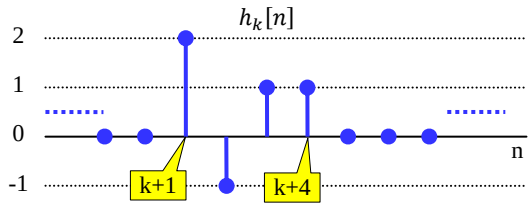
evaluate the overall impulse response of the cascaded system

$$x[n] \longrightarrow z[n].$$

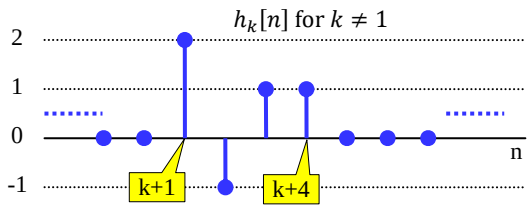
Problem 22.

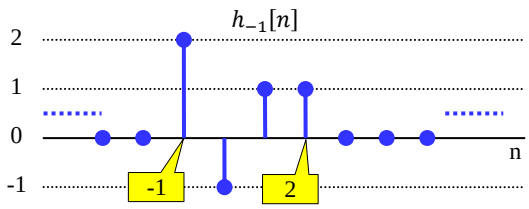
The response of a linear system to a shifted unit impulse $\delta[n-k]$ is called $h_k[n]$.

a) Derive an expression for the system output $[n]$.



(a)





(b)

Figure P22

b) If $h_k[n]$. is given as in Fig. (a), determine whether the corresponding linear system is causal, time-invariant and stable,

c) Repeat part (b) for $h_k[n]$.given as in Fig.(b).

Problem 23.

Determine whether each one of the following statements is TRUE or FALSE and fully justify your answer (i.e. prove if true and disprove or give a counter example if false.) :

i) $y(t) = x(t) * h(t)$ implies $x(t-t_0) * h(t-t_0) = y(t-2t_0)$.

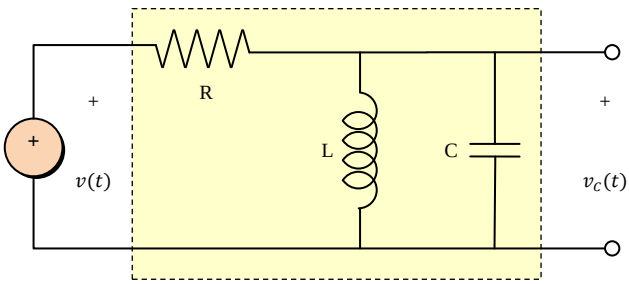
ii) When a linear, time-invariant continuous-time system is excited by a periodic input, the resulting output is also periodic.

iii) $y[n] = x[n] * h[n]$ implies $y[-n] = x[n] * h[-n]$.

iv) The system described by
$x(t) \longrightarrow y(t) = \{x(t)u(t)\} * \{x(-t)u(t)\}$ is causal.

Problem 24.

The circuit shown below is initially at rest. Find a differential equation that expresses the capacitor voltage $v_c(t)$ in terms of the excitation $v(t)$ and from the differential equation find the impulse response $h(t)$.



R=L=C=1

Figure P24

Problem 25.

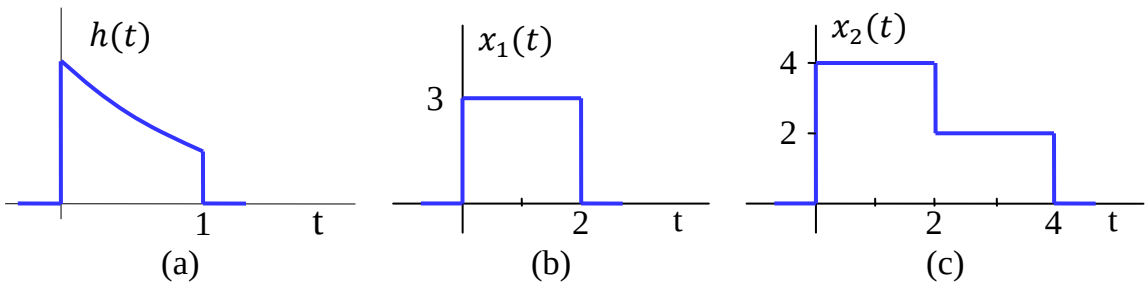


Figure P25

A LTI system has the impulse response

$$h(t) = \begin{cases} e^{-t}, & \text{for } 0 \le t < 1 \\ 0, & \text{elsewhere} \end{cases}.$$

(See figure (a))

a) Find the output $y_1(t)$ corresponding to an input

$$x_1(t) = \begin{cases} 3 & 0 \le t < 2 \\ 0 & \text{elsewhere} \end{cases}.$$

(See figure (b))

b) Express the output $y_2(t)$ for the input

$$x_2(t) = \begin{cases} 4 & 0 \le t < 2 \\ -1 & 2 \le t < 4 \\ 0 & \text{elsewhere} \end{cases}$$

in terms of $y_1(t)$ of part (a). (See figure (c))

c) Evaluate $x_1(t-1) * h(t+1)$ using the result of part (a).

Problem 26.

a) If $x(t) = 0$ for $|t| > T_1$ and $h(t) = 0$ for $|t| > T_2$, then $x(t) * h(t) = 0$ for $|t| > T_3$ for some positive number T_3. Express T_3 in terms of T_1 and T_1.

b) A discrete-time system has input $x[n]$, output $y[n]$ and impulse response $h[n]$. If $h[n]$ is known to be zero outside the interval $N_0 < n < N_1$ and $x[n]$ is known to be zero outside the interval $N_2 < n < N_3$, then the output $y[n]$ is zero everywhere outside the interval $N_4 < n < N_5$.

i) Determine N_4 and N_5 in terms of N_0, N_1, N_2, and N_3.

ii) If the interval $N_0 < n < N_1$ is of length M_h, $N_2 < n < N_3$ is of length M_x and $N_4 < n < N_5$ is of length M_y, express M_y, in terms of M_x and M_h.